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Assessment Paper AY 2024 Semester 2 E211 ESE Main (1)

Section A (12 Questions)

Q1. At a specialist clinic, patients arrive according to fixed appointment times. In this queuing system, the population source would be _____ and the arrival pattern would be _____.

fixed, random

infinite, non-random

finite, non-random

finite, Poisson

Q2. Which of the following methods is used to solve a Maximum Flow problem?

Ford Fulkerson's algorithm

Clarke and Wright savings method

Distribution model

Additive Decomposition method

Q3. Referring to Figure 1a, what type of correlation can be seen between the 2 variables in the chart?

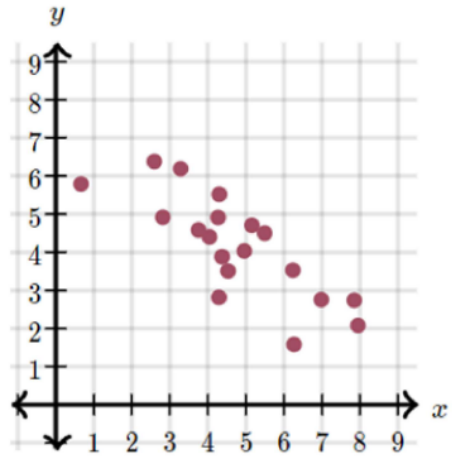

Figure 1a: A Time Series Chart

Positive correlation

Negative correlation

Random correlation

No correlation

Q4. Find a routing solution by applying the Nearest Neighbour Method to the network in Table 1. **Assume starting node is B.**

Table 1: Travel Distance between Nodes

	A	B	C	D
A		8	5	13
B	8		11	9
C	5	11		14
D	13	9	14	

A – C – B – D

B – A – C – D – B

B – C – B – D – A

B – D – A – C – B

Q5. Which one of the following is **NOT** a pitfall when conducting a simulation study?

Failing to define the objectives and scope of the simulation study clearly

Expecting the simulation model to automatically provide an optimal solution to the problem

Neglecting to validate the simulation model before running experiments

Gathering detailed data on the system through time studies and interviews

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Q6. Referring to the statistics obtained from FlexSim simulation (Figure 1b), how many items have passed through Queue1?

Output

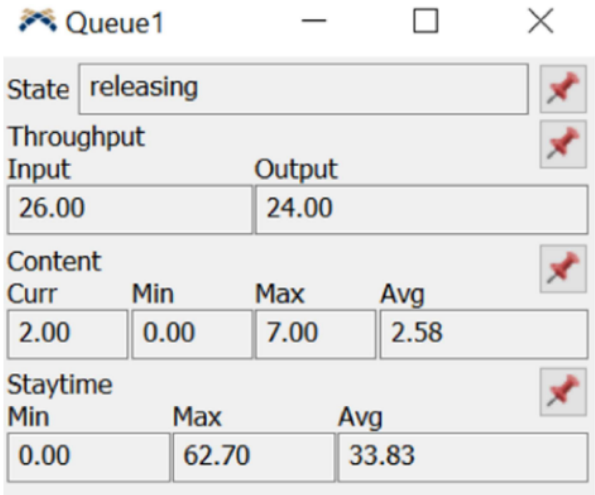


Figure 1b: Queue1 Statistics from FlexSim

7

2

26

24

Throughput: How many items have entered (Input) and exit (Output)

Q7. Referring to the menu bar (Figure 1c) in FlexSim software, what will happen if you uncheck 'Repeat Random Streams' under Statistics?

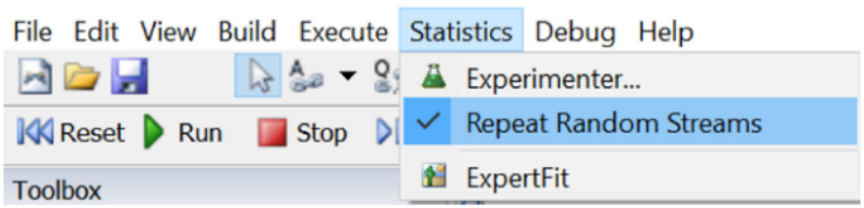


Figure 1c: FlexSim Menu

Results obtained will be different in each simulation run.

There will be no randomness in the simulation.

It will affect Experimenter results.

Simulation results will become more consistent.

Q8. *Show your workings clearly.*

An apparel manufacturer ships products from its warehouse to 6 other locations in the country. The distribution network for its products is shown in Figure 2a. The nodes represent locations and values on the arcs represent distances between locations.

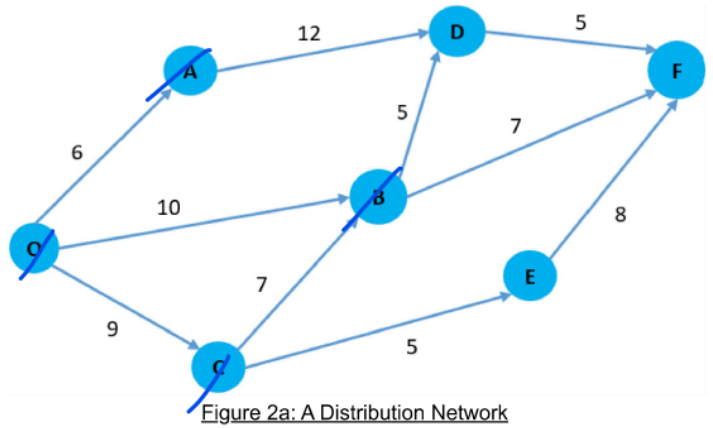


Figure 2a: A Distribution Network

Q8a. Table 2a shows the partial workings to find the shortest path from the warehouse (node O) to each of the other locations (Nodes A to F).

Table 2a: Partial Workings for Shortest Path Problem

Iteration	Solution Set	Arcs	Total Distance
Iteration 1	O	O → A O → B O → C	6 10 9
Iteration 2	O A (6)	O → B O → C A → D	10 9 18
Iteration 3	O A (6) C (9)	O → B A → D C → B C → E	10 18 **V** 16 14
Iteration 4	O A (6) C (9) B (10)	A → D C → E B → D **W1**	18 14 15 **W2** 17
Iteration 5	...	...	...

Q8a(i). Calculate the value of **V** in Table 2a.

V = 9+7 =16

Q8a(ii). In Iteration 4, what is the arc represented by **W1**? State the corresponding Total Distance represented by **W2**.

W1: B ->F
W2 = 10 + 7 = 17

Q8a(iii). Which node should be added into the solution set in **Iteration 5**? Explain.

Node E as it has the next shortest distance of 14.

Q8a(iv). State the shortest path from node O to node D and the total distance of the shortest path from O to D.

O -> B -> D. Total distance = 10 + 5 = 15

Q8b. Figure 2b shows the distribution network of a product. The value at each arc represents the **shipping capacity** along each route.

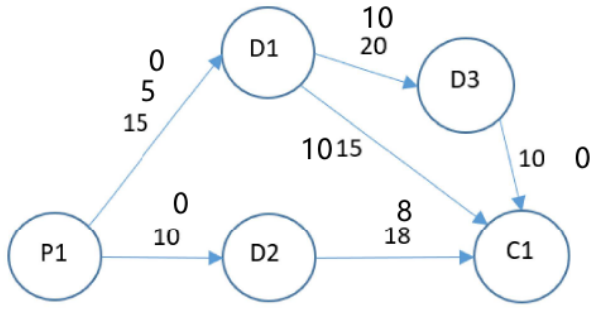


Figure 2b: Product Distribution Network

By following the below paths sequentially, state the flow quantity through each path and determine the maximum flow quantity that can be shipped from P1 to C1.

P1 -> D1 -> D3 -> C1: 10

P1 -> D2 -> C1: 10

P1 -> D1 -> C1: 5

Total maximum flow quantity = 25

Q8c. A distributor ships a new product from a warehouse at Node A. Customers are located at Nodes D & E. The distribution network for shipping the product is shown in Figure 2c where the values on the arcs represent unit shipping costs. The manufacturer wants to determine the lowest cost shipping plan by formulating a Linear Programming (LP) model to solve the Minimum Cost Flow problem.

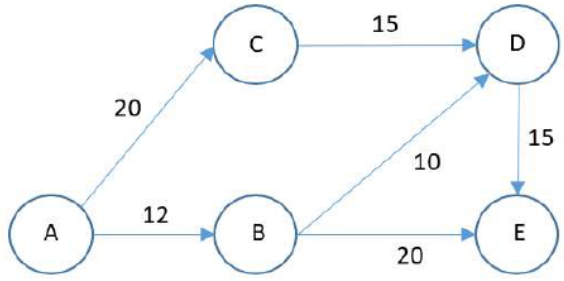


Figure 2c: Distribution Network

The decision variable Xij is defined as the quantity of products to ship from node i to node j (i, j = A, B, C, D or E).

Given:

- Node A can supply up to 300 units
- Nodes D and E have demands of 200 and 100 units respectively
- Nodes B and C are transhipment hubs with zero supply and demand

Q8c(i). How many decision variables will the LP model contain for the Minimum Cost Flow model in Figure 2c?

6

Q8c(ii).

Using the decision variables defined above, formulate the following constraints: 1. Net flow constraint for **Node D**.
2. Maximum shipping capacity from Node B to E is 80.
3. Net flow constraint for **Node A if the demand at Node D is reduced from 200 to 150**.

(i) Node D: XDE - XCD - XBD = -200 (Flow out - Flow in = Net Flow Constraint)
(ii) XBE <= 80
(iii) Node A: XAB + XAC < 300 Node A: XAB + XAC <= 300

Q8d. A product is to be delivered from Depot (A) to a number of stores (B, C, D and E). A truck will leave the Depot (A), visit each store once and return to the Depot after delivery. Travel distances between locations are given in Table 2b.

Table 2b: Travel Distances Between Locations

From	To (Travel distance in km)				
	A	B	C	D	E
A	-	18	9	23	15
B	18	-	12	17	14
C	9	12	-	10	22
D	23	17	10	-	11
E	15	14	22	11	-

Clark and Wright savings method is used to plan the delivery route. Table 2c shows the **Savings** in distance (km) between 2 stores ranked in descending order.

Table 2c: Savings Ranked in Descending Order

Rank	Savings	From	To	
1	27	D	E	A-D-E-A
2	24	B	D	A-B-D-E-A
3	22	C	D	-
4	19	B	E	-
5	16	C	E	A-B-D-E-C-A
6	?	B	C	-

Q8d(i). Calculate the savings between nodes B and C.

Savings = 18+9-12 = 15

Q8d(ii). Referring to Table 2c, at Rank 1, a sub-tour of A-D-E-A is constructed. What step should be carried out next at Rank 2?

Join B and D as D is adjacent to A.

Q8d(iii). State the complete route based on Clark and Wrights Savings Method and calculate the total travel distance of the route.

Complete route: A-B-D-E-C-A

Total travel distance: 18+17+11+22+9 = 77km

Q9. *Show your workings clearly and round off your answer to **2 decimal places** where necessary.*

Katie works in a company that sells health and wellness products. Her role in the company is sales and demand planning.

Q9a. Sales of an existing air purifier has been stagnating recently. To predict customer demand for the next period, Katie collected past sales data (Table 3a) to carry out a forecast.

Table 3a: Historical Sales from Past 6 Months

Month	Sales Figure (Units)
1	80
2	85
3	72
4	77
5	69
6	79

Q9a(i). Calculate the forecast for Month 7 sales using **3-month** moving average.

Month 7 sales = (77+69+79)/3 = 75

Q9a(ii). Calculate the forecast for Month 7 sales using Exponential Smoothing method. Assume smoothing constant is 0.4 and forecast for Month 6 sales is 74.16.

Forecast (month 7) = (0.4*79)+(1-0.4)*74.16 = 76.10

Q9a(iii). Which of the above forecast methods is more responsive to recent changes in the sales data? Explain briefly.

Weighted Moving Average (WMA) as WMA is more responsive to current trends compared with moving average, which weighs all points equally.

Q9a(iv). A colleague suggested that Katie should use Holt's method for the above data forecasting. Do you agree? Explain why.

I do not agree as we do not have Lt and Tt (forecast components) to calculate.

(Explanation: No trend in data.)

Q9a(v). The company has developed a prototype of a new air purifier. Explain how grassroots forecasting method can help predict the demand for the new product.

Provides a lot of detailed knowledge such as insights into regional trends and individual customer needs.

Q9b. Katie developed an Excel template for Multiplicative Decomposition Method (Table 3b) to help her forecast the demand for a seasonal product.

Table 3b: Workings of Multiplicative Decomposition Method

Month/ Week	Actual Value (Y)	Time Period (X)	Moving Average	Centred Moving Average	Seasonal Ratio	Seasonal Index	Deseasonalized value	Deseasonalized forecast	Seasonalized forecast
1/1	742	1				1.627	455.915	462.396	752.547
1/2	467	2	521.75			0.880	530.959	489.064	430.151
1/3	378	3	568	544.875	0.694	0.701	539.496	515.732	361.349
1/4	500	4	576.25	572.125	0.874	0.792	631.066	542.400	429.749
2/1	927	5	(A)		1.589	1.627	569.586	569.068	926.156
2/2	500	6		594.25	0.841	0.880	568.479	595.736	523.974
2/3	434	7	629.5	613.875	0.707	(C)	619.422	622.404	436.089
2/4	532	8	651.25	640.375	0.831	0.792	671.454	649.072	514.266
3/1	1052	9	661.75	656.5	1.602	1.627	646.392	675.740	1099.765
3/2	587	10	667.5	664.625	(B)	0.880	667.394	702.408	617.796
3/3	476	11	715	691.25	0.689	0.701	679.366	729.076	510.830
3/4	555	12	729.5	722.25	0.768	0.792	(D)	755.744	598.784
4/1	1242	13	752.75	741.125	1.676	1.627	763.135	782.412	1273.373
4/2	645	14	764	758.375	0.851	0.880	733.338	809.080	711.619
4/3	569	15	827.75	795.875	0.715	0.701	812.099	835.749	585.570
4/4	600	16	891.5	859.625	0.698	0.792	757.279	862.417	683.301
5/1	1497	17	927.25	909.375	1.646	1.627	919.818	889.085	1446.982
5/2	900	18	977.25	952.25	0.945	0.880	1023.262	915.753	805.441
5/3	712	19				0.701	1016.194	942.421	660.310
5/4	800	20				0.792	1009.706	969.089	767.819
				Intercept for 'Deseasonalized Value'			435.727		
				Slope for 'Deseasonalized Value'			26.668		

Q9b(i). Calculate the Moving Average labelled as **(A)** in Table 3b.

A: (500+927+500+434)/4 = 590.25

Q9b(ii). Calculate the Seasonal Ratio labelled as **(B)** in Table 3b.

B: 587/664.625 = 0.883

Q9b(iii). State the Seasonal Index labelled as **(C)** in Table 3b.

0.701

Q9b(iv). Calculate the Deseasonalized Value labelled as **(D)** in Table 3b.

D: Actual Value (Y) / Seasonal Index = 555/0.792 = 700.758

Q9b(v). Write down the equation of the regression line used to calculate the Deseasonalized Forecast values.

Y = 26.668x + 435.727

Q9b(vi). Calculate the Seasonalized Forecast for **Month 6 Week 2**.

M6 W2 Deseasonalized Forecast = 26.668(22)+435.727 = 1022.423
Seasonal Index = 0.880
Seasonalized Forecast = 1022.423*0.880 = 899.732

Q9b(vii). Calculate the Absolute Percent Error for the forecast in **Month 4 Week 4**.

Absolute deviation (AD) = |600-683.301| = 83.301
Absolute percent error (APE) = 83.301/600 = 0.1388 = 13.88%

Q10. *Show your workings clearly*. ***Round off your answers to 2 decimal places where necessary.***

In a large distribution centre, packages go through labelling before they are shipped. The labelling process is currently done manually by **8 workers**.

A recent study found that packages arrive at the labelling station according to a Poisson process at an average rate of 750 packages per hour. Each package takes an average of 30 seconds to label, and the manual labelling time follows a negative exponential distribution.

Q10a. Calculate the arrival rate of the packages (in per minute) at the labelling station.

Arrival rate = 750/60 = 12.5 per min

Q10b. Calculate the minimum number of workers required at the labelling station for the queuing system to be in a steady state condition.

Service rate = 60/30 = 2 per min
Steady state equilibrium holds when arrival rate < k(service rate) [12.5 <k(2)]
Minimum k > arrival rate/service rate > 12.5/2 > 6.25
Minimum number of workers required, k = 7

Q10c. Given that 8 (value of k) workers are present and n is the number of products in the system in Table 4, calculate the probability that **at least 5 of the workers are busy**.

Table 4: Partial Workings of M/M/8 Queuing Model

n	Probability of n packages in queuing system Pn	
	$\frac{(\lambda/\mu)^n}{n!}P_0$ for $0 \le n \le k$	$\frac{(\lambda/\mu)^n}{k!k^{n-k}}P_0$ for $n > k$
0	0.002	0.747
1	0.011	0.583
2	0.035	0.456
3	0.073	0.356
4	0.114	0.278
5	0.143	0.217
6	0.149	0.170
7	0.133	0.133
8	0.104	0.104
9	0.072	0.081
10	0.045	0.063

P(n>=5) = 1 - P(n<=4)
= 1 - 0.002 - 0.011 - 0.035 - 0.073 - 0.114
= 0.77 (2dp)

Q10d. The labelling supervisor has just purchased an automated labelling machine. The machine performs labelling at a constant time of 10 seconds for every package. If the machine must achieve a utilization of 85%, determine how many packages per hour should be sent to the labelling machine.

Find arrival rate

Arrival rate = 12.5 per min
NEW Service rate = 60/10 = 6 per min

0.85 = Arrival rate/(No of ~~p~~ackages*Service rate)

0.85 = 12.5/(No of packages*6)
0.85(No of packages*6) = 12.5
No of packages*6 = 12.5/0.85 = 14.706
Hence, No of packages (per min) = 14.706/6 = 2.45
No of package (per hour) = 2.45*60 = 147

Service rate = 60/10 = 6 per min
Utilization = Arrival rate/Service rate
0.85 = Arrival rate/6
Arrival rate (in h) = 0.85*6*60 = 306 per hour

Q10e. Assuming the arrival of the packages at the labelling machine follows a Poisson process,

i) Identify a suitable queuing model for the queuing system at the labelling machine.

ii) State 2 assumptions of the identified queuing model.

i) M/M/k X M/D/1

ii) Input source: Infinite population; no balks, no reneging
Arrival process: Number of arrivals follows Poisson distribution

Q10f. On a particular day, the arrival rate of packages to the labelling machine is 4.8 per minute (labelling time is 10 seconds). Calculate:

i) The average number of packages waiting in the queue, Lq

ii) The average time that the packages wait in the queue, Wq in minutes

Arrival rate = 4.8 per min, Service rate = 60/10 = 6 per min

i) Lq = 2.30*10^-7 Lq = 1.6

ii) Wq = 4.80*10^-8

i. Utilization = 4.8/6 = 0.8
Lq = 0.8^2/(2*(1-0.8)) = 1.6

ii. Wq = Lq/Arrival rate = 1.6/4.8 = 0.33

Q11. Figure 5a shows a simulation snapshot of a goods receiving process modelled in FlexSim software. Incoming items are first unpacked before they are sent to QC. If an item fails QC, it will be sent to Quarantine. If it passes QC, it will go to Stock In.

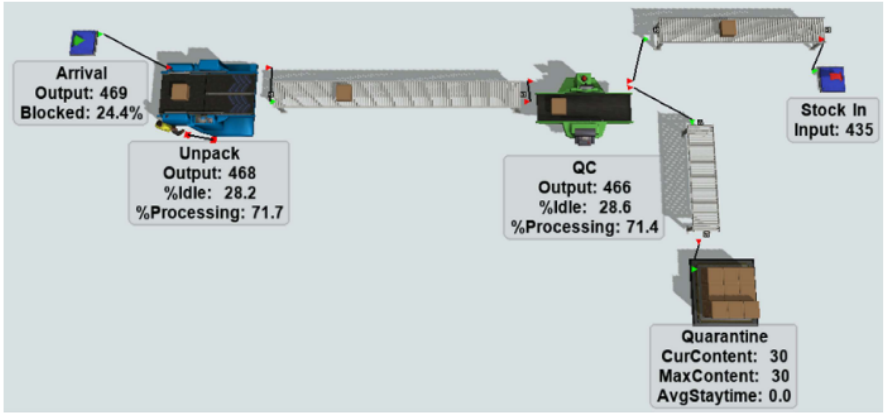


Figure 5a: FlexSim Model of Goods Receiving

Q11a. Referring to Figure 5a, state which FlexSim 3D object is used to model the following:

Stock In: sink

QC: processor

Q11b. Referring to Figure 5a, state the following statistics:

Number of items in Quarantine = 30 (CurContent)

Utilization of Unpack = 71.7 (%processing)

Q11c. It is known that 6% of incoming items will need to be quarantined. At the QC object, the 'Send to Port' option is set to 'By Percentage' (shown in Figure 5b). Specify the values of A and B in the input box.

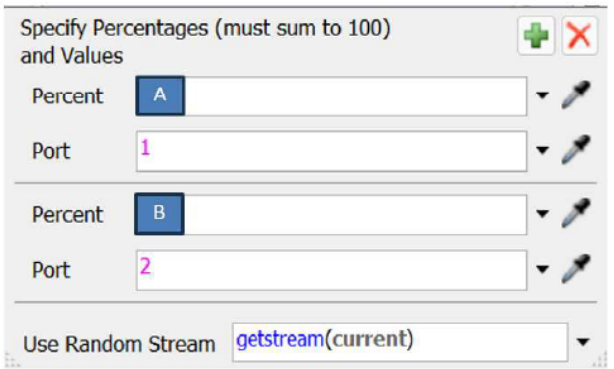


Figure 5b: By Percentage Input Box

A: 94

B: 6

Q11d. Figure 5c shows the settings of the 'On Exit' trigger for the QC object. Explain what this trigger does.

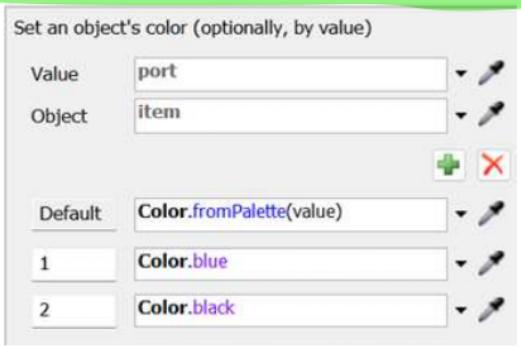


Figure 5c: On Exit Trigger in QC Object

If item pass QC, item color will be blue when going to stock in.
If item fails QC, item color will be black when going to quarantine.

Q12. Jack is using a FlexSim Process Flow model to study a customer service process at a call center. Figure 6a shows a snapshot of the simulation.

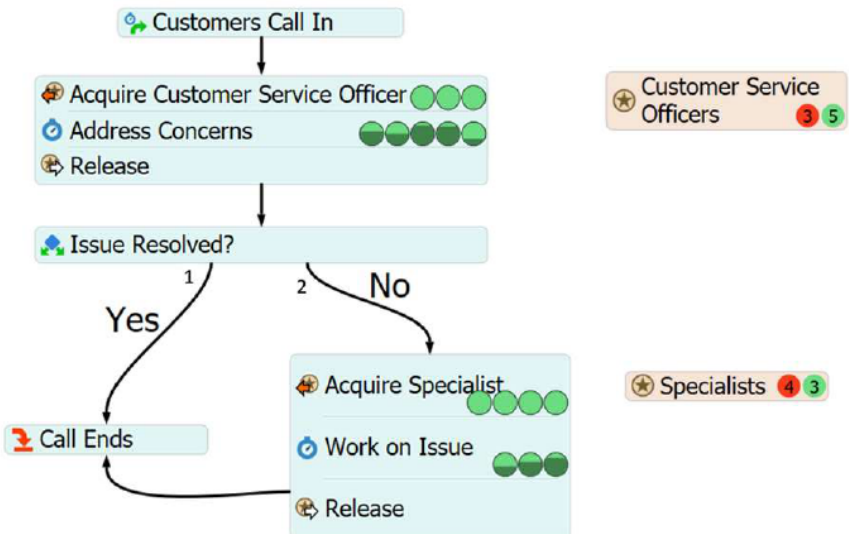


Figure 6a: FlexSim Process Flow Simulation

Q12a.

How many customers are currently waiting for a Specialist?

4 (look at the red icon "specialist")

ii) How many Customer Service Officers are there?

8 X 5

iii) Identify the FlexSim Process Flow activity is used to model 'Issue Resolved?'.

Answer the following questions:i) decide

Q12b. Referring to Figure 6b, what values should Jack input for X and Y if 30% of the calls require a specialist to intervene?

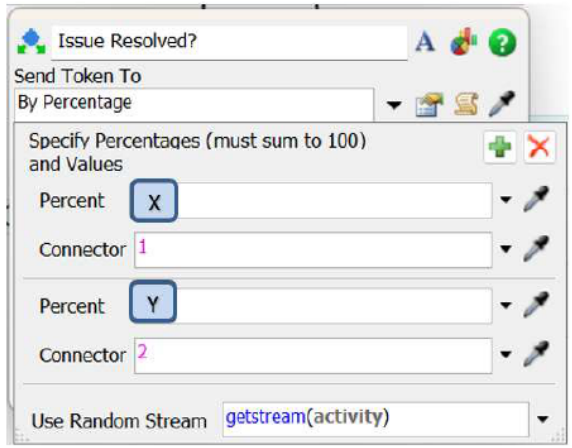
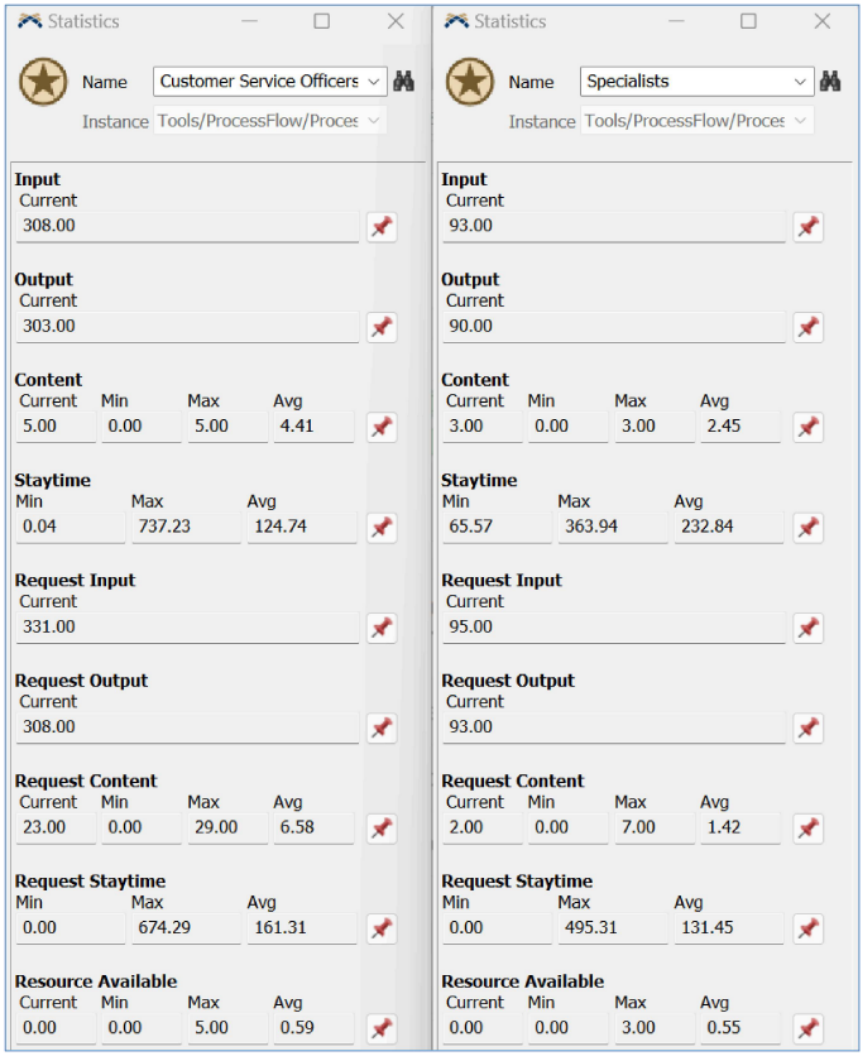


Figure 6b: Activity Properties

X = 70

Y = 30

Q12c. Figure 6c shows the statistics for Customer Service Officers and Specialists at the end of a simulation run.



Statistics

Name: Customer Service Officers
Instance: Tools/ProcessFlow/Proces

	Current	Min	Max	Avg
Input	308.00			
Output	303.00			
Content	5.00	0.00	5.00	4.41
Staytime		0.04	737.23	124.74
Request Input	331.00			
Request Output	308.00			
Request Content	23.00	0.00	29.00	6.58
Request Staytime		0.00	674.29	161.31
Resource Available	0.00	0.00	5.00	0.59

Statistics

Name: Specialists
Instance: Tools/ProcessFlow/Proces

	Current	Min	Max	Avg
Input	93.00			
Output	90.00			
Content	3.00	0.00	3.00	2.45
Staytime		65.57	363.94	232.84
Request Input	95.00			
Request Output	93.00			
Request Content	2.00	0.00	7.00	1.42
Request Staytime		0.00	495.31	131.45
Resource Available	0.00	0.00	3.00	0.55

Figure 6c: Statistics of Work Station and QC Station

[redacted]

[Request content]
CS officers = 6.58, Specialists = 1.42

Q12c(i). Determine the bottleneck of the process by comparing the average queue length for the 2 resources.

[We look at avg content]: Avg queue length (CS officers) = 4.41, Avg queue length (specialists) = 2.45

Hence bottleneck will be Customer Service Officers

Q12c(ii). Calculate the utilization of the Specialists.

Average utilization = Avg content/Max resource = 2.45/3 = 0.817 = 81.7%

Q12d. Someone suggested that Jack should build a 3D model to study the customer service process. Do you agree? Explain why.

I don't agree as customers will be calling in digitally, if they will walk in, 3D model will make sense as we can get to see how long is the queue length.

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